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DEPARTMENT: CSE

C - MINI PROJECT DOCUMENTATION

TITLE: INTERACTIVE CALENDAR APPLICATION WITH NOTE MANAGEMENT AND FILE EXPORT

PROBLEM STATEMENT:

Managing schedules, tracking important events, and planning effectively for long-term goals can be challenging without a systematic and accessible solution. Individuals often struggle to keep track of events, deadlines, and preparation schedules, resulting in missed opportunities and inefficiencies. There is a need for a user-friendly application that not only enables users to create and manage personalized calendars with event annotations and detailed notes but also provides options to save, print, and visualize these plans for easy accessibility and effective time management.

AIM:

To develop a calendar application in C that enables users to view and manage monthly or yearly calendars, add notes for specific dates, and save calendars or notes to text files for future reference, providing a comprehensive and user-friendly utility.

ALGORITHM:

1. **Initialization**:
    - Prompt the user for the year of the calendar.
    - Initialize variables for menu options and input handling.
2. **Display Menu**:
    - Show a menu with options such as viewing calendars, adding notes, saving files, and exiting.
3. **Input Handling**:
    - Accept the user's choice and validate the input.
    - Call appropriate functions based on the choice.
4. **View Calendar**:
    - For a full year: Loop through months (1-12) and display each month using the logic for day calculations.
    - For a specific month: Display the selected month's calendar.
5. **Add Notes**:
    - Prompt for the date and note details.
    - Save notes to an internal data structure or a file for persistence.
6. **Save to File**:
    - Ask the user if they want to save a specific month's calendar or the full year.
    - Check for file overwrites and confirm saving.
7. **Exit**:
    - Provide an option to quit the application cleanly.

TOOLS USED:

1. **Programming Language**: C
2. **Libraries**:
    - <stdio.h>: For input/output operations.
    - <stdlib.h>: For utility functions.

- <time.h>: For handling dates.

3. **Data Storage**: Files (to save calendar and notes).

INPUT:

1. Year for the calendar.
2. Menu choice for the desired operation.
3. Specific month (if applicable).
4. Note details (date and text).
5. File name for saving data.

PROGRAM:

```
#include <stdio.h>

#include <string.h>

#define RESET "\x1B[0m"

#define GREEN "\x1B[32m"

#define YELLOW "\x1B[33m"

// Array to store notes for each date

char notes[12][31][100] = {{{0}}};

// Function that returns the index of the day for date DD/MM/YYYY

int dayNumber(int day, int month, int year) {

    static int t[] = {0, 3, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4};

    year -= month < 3;

    return (year + year / 4 - year / 100 + year / 400 + t[month - 1] + day) % 7;
```

```
}
// Function to return the name of the month for the given month number
char *getMonthName(int monthNumber) {
    static char *months[] = {"January", "February", "March", "April", "May",
                             "June", "July", "August", "September", "October",
                             "November", "December"};
    return months[monthNumber];
}
// Function to return the number of days in a month
int numberOfDays(int monthNumber, int year) {
    if (monthNumber == 1) { // February
        if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
            return 29;
        else
            return 28;
    }
    if (monthNumber == 3 || monthNumber == 5 || monthNumber == 8 || monthNumber == 10)
        return 30;
    return 31;
}
// Function to print the calendar for a specific month
void printMonthCalendar(int year, int month) {
    printf("\n ------------%s-------------\n", getMonthName(month - 1));
```

```
printf(" Sun    Mon    Tue    Wed    Thu    Fri    Sat\n");
int days = numberOfDays(month - 1, year);
int current = dayNumber(1, month, year);
int k;
for (k = 0; k < current; k++)
    printf("     ");
for (int j = 1; j <= days; j++) {
    if (k % 7 == 0) { // Sunday in GREEN
        if (notes[month - 1][j - 1][0] != '\0') // Notes in YELLOW
            printf(YELLOW "%5d" RESET, j);
        else
            printf(GREEN "%5d" RESET, j);
    } else if (notes[month - 1][j - 1][0] != '\0') { // Notes in YELLOW
        printf(YELLOW "%5d" RESET, j);
    } else {
        printf("%5d", j);
    }
    if (++k > 6) {
        k = 0;
        printf("\n");
    }
}
if (k)
```

```
        printf("\n");
    // Print notes for the month
    printf("\nNotes for %s:\n", getMonthName(month - 1));
    for (int day = 0; day < 31; day++) {
        if (notes[month - 1][day][0] != '\0') {
            printf("  %2d: %s\n", day + 1, notes[month - 1][day]);
        }
    }
}
// Function to save the calendar for selected months to a file
void saveSelectedMonthsToFile(int year, int *months, int count) {
    // Create a filename for the selected months
    char filename[100] = "calendar_";
    for (int i = 0; i < count; i++) {
        char monthName[20];
        sprintf(monthName, "%s_", getMonthName(months[i] - 1));
        strcat(filename, monthName);
    }
    sprintf(filename + strlen(filename) - 1, "_%d.txt", year); // Add year to the filename
    FILE *file = fopen(filename, "w");
    if (file == NULL) {
        printf("Error: Unable to save file.\n");
        return;
```

```
}
fprintf(file, "     Calendar - Selected Months of %d\n\n", year);
for (int i = 0; i < count; i++) {
    int month = months[i];
    fprintf(file, "\n ------------%s-------------\n", getMonthName(month - 1));
    fprintf(file, " Sun Mon Tue Wed Thu Fri Sat\n");
    int days = numberOfDays(month - 1, year);
    int current = dayNumber(1, month, year);
    int k;
    for (k = 0; k < current; k++)
        fprintf(file, "     ");
    for (int j = 1; j <= days; j++) {
        if (k % 7 == 0) { // Sunday
            if (notes[month - 1][j - 1][0] != '\0') // Notes
                fprintf(file, "%5d*", j);
            else
                fprintf(file, "%5d", j);
        } else if (notes[month - 1][j - 1][0] != '\0') { // Notes
            fprintf(file, "%5d*", j);
        } else {
            fprintf(file, "%5d", j);
        }
        if (++k > 6) {
```

```
                k = 0;
                fprintf(file, "\n");
            }
        }
        if (k)
            fprintf(file, "\n");
        // Append notes for the current month
        fprintf(file, "\nNotes for %s:\n", getMonthName(month - 1));
        for (int day = 0; day < 31; day++) {
            if (notes[month - 1][day][0] != '\0') {
                fprintf(file, "  %2d: %s\n", day + 1, notes[month - 1][day]);
            }
        }
    }
    fclose(file);
    printf("Calendar for selected months saved to %s!\n", filename);
}
// Function to add notes
void addNote() {
    int month, day;
    char note[100];
    printf("Enter month (1-12): ");
    scanf("%d", &month);
```

```
    printf("Enter day (1-31): ");
    scanf("%d", &day);
    printf("Enter note: ");
    scanf(" %[^\n]s", note);
    strcpy(notes[month - 1][day - 1], note);
    printf("Note added successfully!\n");
}
// Function to save full calendar
void saveFullYearToFile(int year) {
    char filename[50];
    sprintf(filename, "calendar_%d.txt", year);
    FILE *file = fopen(filename, "w");
    if (file == NULL) {
        printf("Error: Unable to save file.\n");
        return;
    }
    fprintf(file, "     Full Year Calendar - %d\n\n", year);
    for (int month = 1; month <= 12; month++) {
        fprintf(file, "\n ------------%s-------------\n", getMonthName(month - 1));
        fprintf(file, " Sun Mon Tue Wed Thu Fri Sat\n");
        int days = numberOfDays(month - 1, year);
        int current = dayNumber(1, month, year);
        int k;
```

```
for (k = 0; k < current; k++)
    fprintf(file, "     ");
for (int j = 1; j <= days; j++) {
    if (k % 7 == 0) { // Sunday
        if (notes[month - 1][j - 1][0] != '\0') // Notes
            fprintf(file, "%5d*", j);
        else
            fprintf(file, "%5d", j);
    } else if (notes[month - 1][j - 1][0] != '\0') { // Notes
        fprintf(file, "%5d*", j);
    } else {
        fprintf(file, "%5d", j);
    }
    if (++k > 6) {
        k = 0;
        fprintf(file, "\n");
    }
}
if (k)
    fprintf(file, "\n");
// Append notes for the current month
fprintf(file, "\nNotes for %s:\n", getMonthName(month - 1));
for (int day = 0; day < 31; day++) {
```

```
            if (notes[month - 1][day][0] != '\0') {
                fprintf(file, "  %2d: %s\n", day + 1, notes[month - 1][day]);
            }
        }
    }
    fclose(file);
    printf("Full year calendar saved to %s!\n", filename);
}
// Main function
int main() {
    int year, choice, month;
    printf("Calendar Application\n");
    printf("Enter the year: ");
    scanf("%d", &year);
    do {
        printf("\nMenu:\n");
        printf("1. Display full year calendar\n");
        printf("2. Display specific month calendar\n");
        printf("3. Add a note\n");
        printf("4. Show notes for a month\n");
        printf("5. Save calendar for specific months\n");
        printf("6. Save full calendar\n");
        printf("7. Exit\n");
```

```
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
    for (int i = 1; i <= 12; i++)
        printMonthCalendar(year, i);
    break;
case 2:
    printf("Enter month (1-12): ");
    scanf("%d", &month);
    printMonthCalendar(year, month);
    break;
case 3:
    addNote();
    break;
case 4:
    printf("Enter month (1-12): ");
    scanf("%d", &month);
    printMonthCalendar(year, month); // Show notes included in month view
    break;
case 5: {
    int count;
    printf("How many months do you want to save? ");
```

```
            scanf("%d", &count);
            int months[count];
            printf("Enter the months (1-12): ");
            for (int i = 0; i < count; i++) {
                scanf("%d", &months[i]);
            }
            saveSelectedMonthsToFile(year, months, count);
            break;
        }
        case 6:
             saveFullYearToFile(year);
                break;
        case 7:
             printf("Exiting...\n");
                 break;
        default:
            printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 7);
    return 0;
}
```

EXECUTION STEPS:

1. **Launch the Application**:
    - Run the program and enter the year.
2. **Menu Interaction**:
    - Choose from options:
        - Display the full calendar.
        - View a specific month.
        - Add notes.
        - Show notes.
        - Save data to a file.
        - Exit the application.
3. **Calendar Viewing**:
    - Display calendars using a structured format with days and dates aligned.
4. **Notes Management**:
    - Add or retrieve notes for specific dates.
5. **Save Data**:
    - Save the selected calendar (monthly or yearly) or notes to a text file.
6. **Exit**:
    - Confirm exit and terminate the program.

OUTPUT:

```
E:\MY FOLDER\C PROJECT\c_p

Calendar Application
Enter the year: 2025

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 1

 ------------January-------------
 Sun    Mon    Tue    Wed    Thu    Fri    Sat
                   1    2    3    4
    5    6    7    8    9   10   11
   12   13   14   15   16   17   18
   19   20   21   22   23   24   25
   26   27   28   29   30   31

Notes for January:

 ------------February------------
 Sun    Mon    Tue    Wed    Thu    Fri    Sat
                                    1
    2    3    4    5    6    7    8
    9   10   11   12   13   14   15
   16   17   18   19   20   21   22
   23   24   25   26   27   28

Notes for February:
```

```
E:\MY FOLDER\C PROJECT\c_p

Notes for December:

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 2
Enter month (1-12): 3

 ------------March-------------
 Sun    Mon    Tue    Wed    Thu    Fri    Sat
                                    1
    2    3    4    5    6    7    8
    9   10   11   12   13   14   15
   16   17   18   19   20   21   22
   23   24   25   26   27   28   29
   30   31

Notes for March:

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice:
```

```
 Sun    Mon    Tue    Wed    Thu    Fri    Sat
                                       1
    2      3      4      5      6      7      8
    9     10     11     12     13     14     15
   16     17     18     19     20     21     22
   23     24     25     26     27     28     29
   30     31

Notes for March:

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 3
Enter month (1-12): 3
Enter day (1-31): 10
Enter note: Its my birthday
Note added successfully!

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice:
```

```
Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 4
Enter month (1-12): 3

 ------------March-------------
 Sun    Mon    Tue    Wed    Thu    Fri    Sat
                                       1
    2      3      4      5      6      7      8
    9     10     11     12     13     14     15
   16     17     18     19     20     21     22
   23     24     25     26     27     28     29
   30     31

Notes for March:
  10: Its my birthday

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice:
```

```
E:\MY FOLDER\C PROJECT\c_p

 Sun    Mon    Tue    Wed    Thu    Fri    Sat
                                       1
    2      3      4      5      6      7      8
    9     10     11     12     13     14     15
   16     17     18     19     20     21     22
   23     24     25     26     27     28     29
   30     31

Notes for March:
  10: Its my birthday

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 5
How many months do you want to save? 2
Enter the months (1-12): 2 3
Calendar for selected months saved to calendar_February_March_2025.txt!

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice:
```

```
E:\MY FOLDER\C PROJECT\c_p

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 5
How many months do you want to save? 2
Enter the months (1-12): 2 3
Calendar for selected months saved to calendar_February_March_2025.txt!

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 6
Full year calendar saved to calendar_2025.txt!

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice:
```

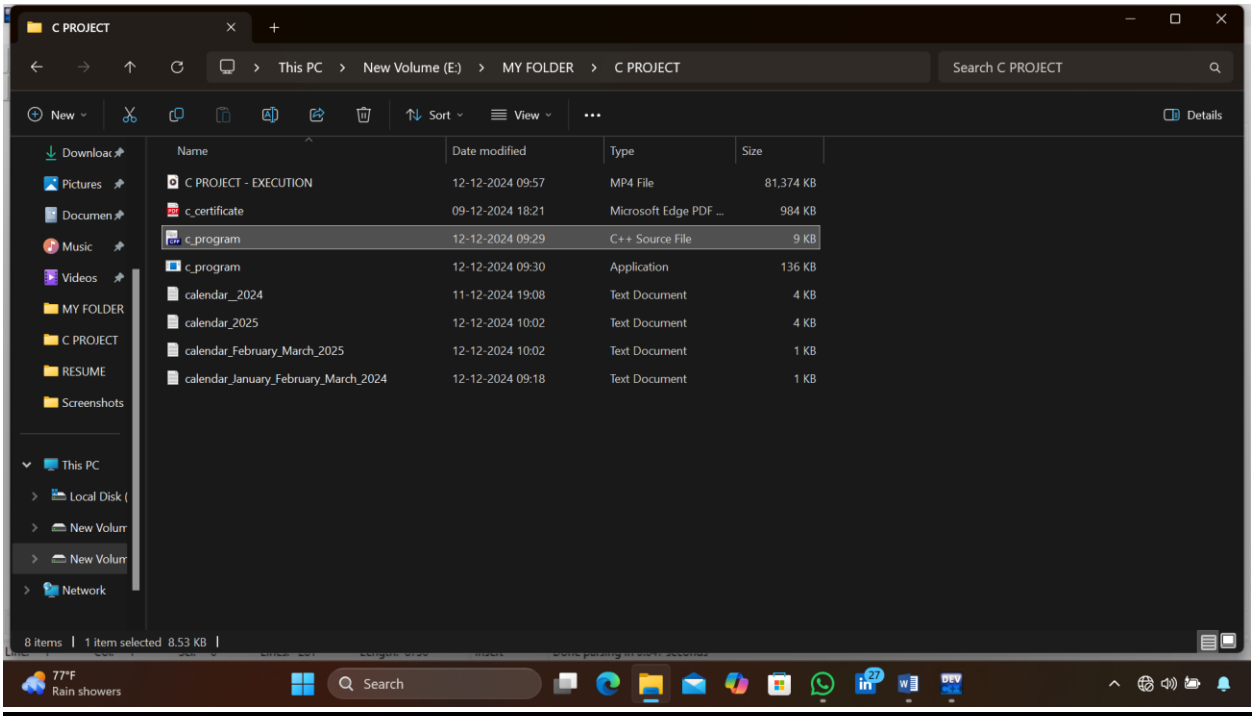
```
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 5
How many months do you want to save? 2
Enter the months (1-12): 2 3
Calendar for selected months saved to calendar_February_March_2025.txt!

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 6
Full year calendar saved to calendar_2025.txt!

Menu:
1. Display full year calendar
2. Display specific month calendar
3. Add a note
4. Show notes for a month
5. Save calendar for specific months
6. Save full calendar
7. Exit
Enter your choice: 7
Exiting...

--------------------------------
Process exited after 475.2 seconds with return value 0
Press any key to continue . . .
```

SAVED CALENDAR:



calendar_February_March_2025.txt

File Edit View

```
     Calendar - Selected Months of 2025


 ------------February-------------
 Sun Mon Tue Wed Thu Fri Sat
                                 1
    2   3   4   5   6   7   8
    9  10  11  12  13  14  15
   16  17  18  19  20  21  22
   23  24  25  26  27  28

Notes for February:

 ------------March-------------
 Sun Mon Tue Wed Thu Fri Sat
                                 1
    2   3   4   5   6   7   8
    9  10*  11  12  13  14  15
   16  17  18  19  20  21  22
   23  24  25  26  27  28  29
   30  31

Notes for March:
  10: Its my day
```

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77°F Rain showers | Search

calendar_February_March_2025.txt | calendar_2025.txt

File Edit View

```
     Full Year Calendar - 2025


 ------------January-------------
 Sun Mon Tue Wed Thu Fri Sat
                   1   2   3   4
    5   6   7   8   9  10  11
   12  13  14  15  16  17  18
   19  20  21  22  23  24  25
   26  27  28  29  30  31

Notes for January:

 ------------February-------------
 Sun Mon Tue Wed Thu Fri Sat
                                 1
    2   3   4   5   6   7   8
    9  10  11  12  13  14  15
   16  17  18  19  20  21  22
   23  24  25  26  27  28

Notes for February:

 ------------March-------------
 Sun Mon Tue Wed Thu Fri Sat
                                 1
    2   3   4   5   6   7   8
    9  10*  11  12  13  14  15
   16  17  18  19  20  21  22
   23  24  25  26  27  28  29
   30  31

Notes for March:
```

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77°F Rain showers | Search

USER CASE:

1. **Event Tracking and Documentation**: The application facilitates the marking and documentation of events or activities for an entire month or year. Users can annotate specific dates with relevant notes, save the calendar, and generate a printable version for maintaining a systematic record, ensuring accessibility and ease of reference.
2. **Planning and Schedule Management**: The application enables users to create comprehensive plans for specific periods, such as a three-month preparation schedule for competitive exams like JEE. Users can add detailed notes to outline the syllabus or tasks to be covered on particular dates. The calendar can be printed and displayed in prominent locations to serve as a constant reminder, aiding in effective time management and goal tracking.

RESULT:

The application effectively displays a well-structured calendar for the specified year or month, complete with weekdays and proper formatting. Users can add, view, and manage notes for specific dates, making it convenient to organize important events. Additionally, the calendar and notes can be saved to a file, ensuring easy access and reference in the future.